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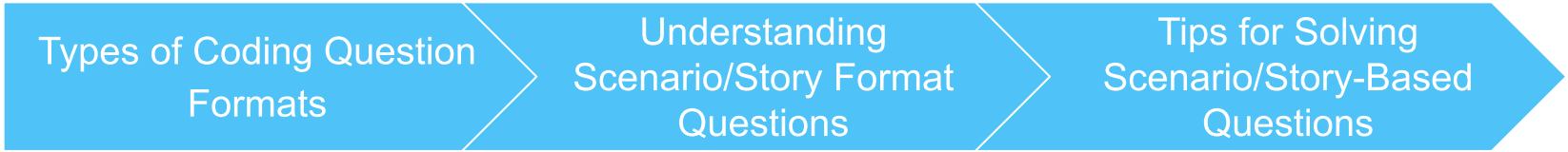
WELCOME

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Understanding Coding Question Formats

Agenda for Today's Session

Understanding Coding Question Formats



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Let’s see some sample Coding Questions





Example - Coding Questions

Example 1

Sarah is teaching mathematics to primary school students and wants to make learning fun.
She plans to create a game to help students understand division by finding all divisors of a given number. The game will display all numbers that can divide the given number evenly, without leaving a remainder. Help Sarah to create this game

Example 2

Write a program to print all factors of a given number

Question

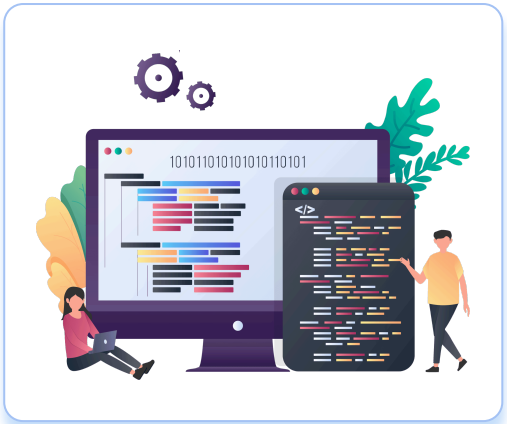


What difference did you observe?



Types of Coding Question Formats

Types of Coding Question Formats



Plain Technical Format



Scenario/Story Format

many more...

Types of Coding Question Formats

Plain Technical Format - Example

Write a program to print all factors of a given number

Types of Coding Question Formats

Plain Technical Format - Example

Write a program to print all factors of a given number

This is the most straightforward format where:

- The **requirement** is clearly stated
- **Technical terms** are directly used
- The expected **output is clear**

Example 1 - Coding Questions

Sarah is teaching mathematics to primary school students and wants to make learning fun

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Types of Coding Question Formats

Game Based Formats - Example

> Your program must destroy the enemy ships by shooting the closest enemy on each turn

The question format presented is a **game-based task** where

- The primary requirement, "destroy the enemy ships," is explicitly mentioned
- The strategy for achieving the goal is: "shooting the closest enemy on each turn"

Project Based Formats - Example

Create a Weather Application that:

1. Fetches weather data from an API
2. Allows user to search by city
3. Displays temperature, humidity, wind speed

Code Optimization Format

Write a function:
def solution(N):

that, given a positive integer N, returns the number of its even factors.

For example, given N = 16, the function should return 3, because among all factors of 16 (1, 2, 4, 8, 16), only three factors (2, 4, 8) are even numbers.
Here's an inefficient solution:

```
def solution(N):
    count = 0
    # Check all numbers from 1 to N
    for i in range(1, N + 1):
        # Check if i is a factor of N
        if N % i == 0:
            # Check if the factor is even
            if i % 2 == 0:
                count += 1
    return count
```

The above solution works correctly but is highly inefficient for large numbers as it:

- Checks every number from 1 to N
- Makes unnecessary iterations
- Has a time complexity of O(N)

Write an efficient algorithm for the following assumptions:

- N is an integer within the range [1..2,147,483,647]

Efficient:

- Takes less time to run
- Uses less no of iterations, etc

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Question

Have you ever practice these type of questions ?



Types of Coding Question Formats

Coding Questions

In technical interviews and coding assessments, you'll frequently encounter these formats

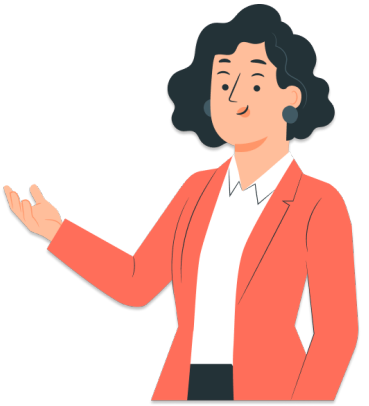


Types of Coding Question Formats

Scenario/Story Format

Many companies deliberately use story-based questions to evaluate

- Your ability to extract technical requirements from real-world scenarios
- Problem-solving approach in practical situations
- Understanding of how programming solutions apply to business context

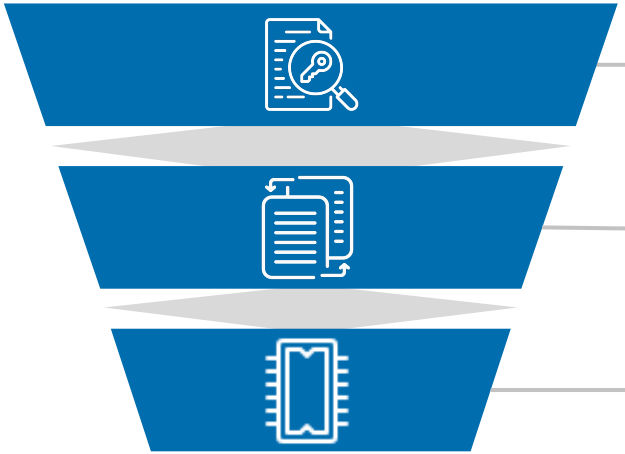


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Understanding

Scenario/Story Format Questions

Scenario/Story Format Questions



1. Identify Key Information
2. Convert to Technical Terms
3. Extract the Core Problem

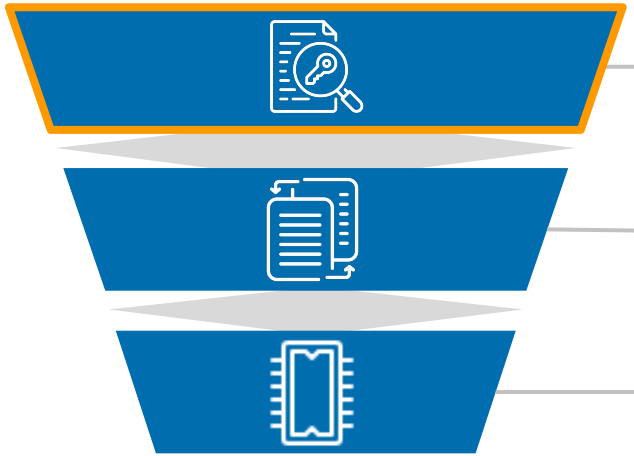
Understanding Scenario/Story Based Question

Example

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She plans to create a game to help students understand division by finding all divisors of a given number. The game will display all numbers that can divide the given number evenly, without leaving a remainder. Help Sarah to create this game

Scenario/Story Format Questions



1. Identify Key Information
2. Convert to Technical Terms
3. Extract the Core Problem

Understanding Scenario/Story Format Questions

Identify Key Information

Identifying Key Information



Main Character: Sarah (Teacher)



Problem Context: Teaching mathematics

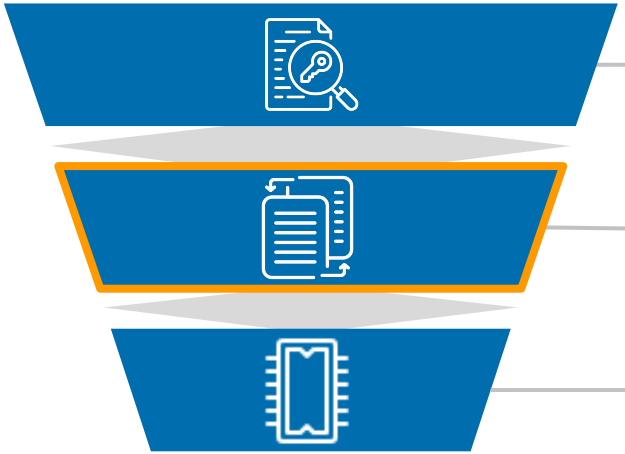


Requirement: Finding numbers that divide evenly



Technical Term: These numbers are called factors

Scenario/Story Format Questions



1. Identify Key Information
2. Convert to Technical Terms
3. Extract the Core Problem

Understanding Scenario/Story Format Questions

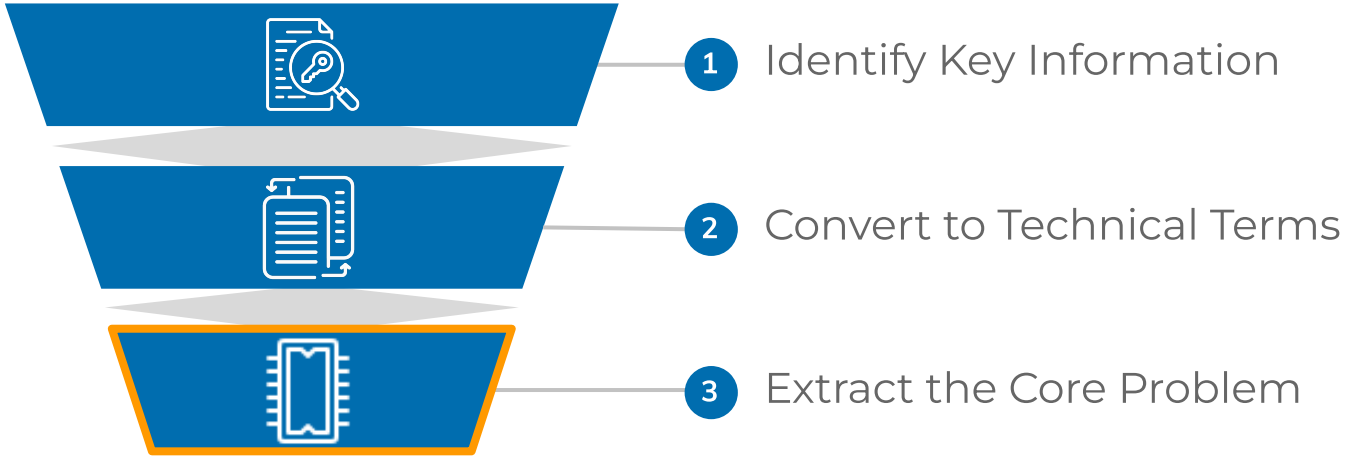
Convert to Technical Terms

1. "numbers that divide evenly" = factors

2. "without any remainder" = division with remainder 0



Scenario/Story Format Questions



Understanding Scenario/Story Format Questions

Extract the Core Problem



Original story → Find factors of a number



Input → A number



Output → List of all its factors

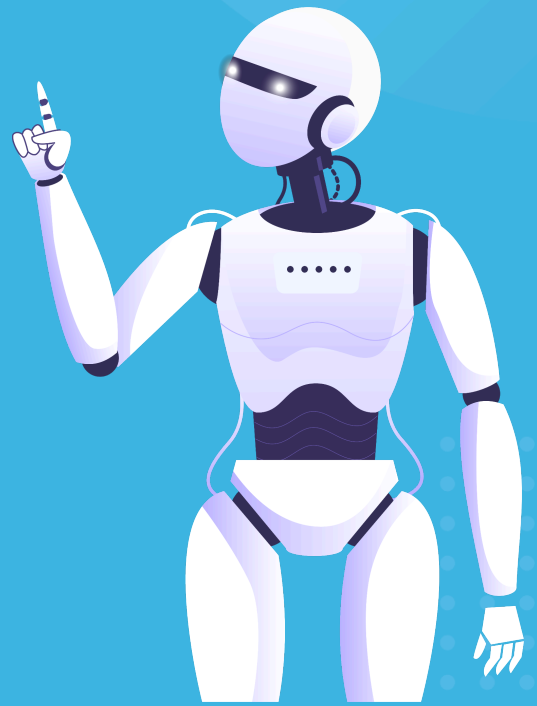
Finding Factors of a Number

Approach

1. Take a number as input
2. Try dividing it with numbers from 1 to the number itself
3. If division gives remainder 0, it's a factor
4. Print the factor

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Write approach using pen/paper before jumping into coding



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Examples

Finding if a Number is Palindrome

without Scenario/Story

Write a program to check if a given number or string is a palindrome. A palindrome reads the same backward as forward

Scenario/Story

A linguist is studying ancient scrolls that are believed to contain hidden messages. These messages are considered authentic only if they read the same forwards and backwards. Create a program to help the linguist verify the authenticity of these messages

Examples

List Sum

without Scenario/Story

Write a program to find two numbers in an array that add up to a given target sum

Scenario/Story

Imagine you're a cashier at a busy store. Customers can use two discount coupons together to get a specific total discount amount. Given a pile of discount coupons with different values, you need to quickly find which two coupons can be combined to reach the exact discount amount the customer wants

Examples

Finding Missing Number

without Scenario/Story

Find the missing number in an array containing n distinct numbers taken from 0 to n

Scenario/Story

In a classroom, students are playing a number game where each student gets a unique number from 0 to n. When the teacher collects the numbers back, she realizes one student forgot to return their number. Help the teacher write a program to figure out which number is missing

Tips for Solving

Scenario/Story Based Questions

Tips for Solving Scenario/Story Based Questions



Read Carefully



Look for Technical Clues



Organize Information

Tips for Solving Scenario/Story Based Questions

Read Carefully



- Highlight important information
- Identify what is given and what is asked

Tips for Solving Scenario/Story Based Questions

Look for Technical Clues

- "divide evenly" → division operation
- "all possible ways" → need to find all solutions



Tips for Solving Scenario/Story Based Questions

Organize Information

- Input requirements
- Expected output
- Any constraints mentioned



Key Takeaways

- Types of Coding Question Formats
- Understanding Scenario/Story Format Questions
- Tips for Solving Scenario/Story Based Questions



Practice



Feedback



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THANK YOU

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ALL THE BEST